



Lesson 8.8 — Experimental Design

Big idea: A well-designed experiment isolates cause from confounds. Randomization, control, replication, and blinding turn a noisy world into a fair test.

Quick review

Observational vs. experimental: observational measures without intervening (association only); experiment applies treatments (cause and effect possible).

Four principles: random assignment, control/comparison group, replication, and blinding.

Confounding variable: something tied up with the treatment that could explain the observed difference.

Placebo effect: response to receiving any treatment. Counter with a placebo control.

Worked example

Worked example. A new fertilizer is applied to half the tomatoes; the new-fertilizer plants happen to be in the sunny part of the field.

Confound: sunlight tangled with treatment. Fix: randomize plants to treatments throughout the field.

Warm-ups

1. Is a poll of patients about their hospital experience an experiment or observational study?
2. What does it mean for an experiment to be “double-blind”?
3. Why include a control group?

Practice

1. Identify a possible confound: studies show coffee drinkers have higher rates of heart disease.
2. Design a simple experiment to test whether brand A or brand B of running shoe results in faster 5K times among runners. Specify treatment, randomization, and controls.
3. Why is random ASSIGNMENT (in an experiment) different from random SAMPLING (in a survey)?

Challenge

1. A teacher believes a new study app improves test scores. She lets students in her advanced class use it; her standard class doesn't. She compares average scores. Critique the design and propose a fix.
2. A drug trial gives the new drug to 200 patients and a placebo to 200 patients. Both groups improve, the new-drug group only slightly more. Why is the placebo group essential to interpret this result?



Answer key — Lesson 8.8

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Observational.
2. Neither the subjects nor the people measuring the response know who got which treatment, removing expectation bias.
3. To compare against a baseline so any effect of the treatment can be measured against natural variability or placebo.

Practice

1. Confound: smoking. Smokers tend to drink more coffee AND have higher heart disease rates, so the apparent coffee effect could be smoking.
2. Recruit say 60 runners. Randomly assign 30 to brand A, 30 to brand B. Each runs a 5K under similar conditions (same course, similar weather, comparable distance). Compare mean times. Optionally crossover later for within-runner comparison.
3. Random sampling: how subjects are chosen FROM the population (gives representativeness). Random assignment: how subjects are split AMONG treatment groups within an experiment (gives causal inference). Both are random, but they answer different questions.

Challenge

1. Students self-selected into classes; the advanced class differs from standard in many ways (motivation, prior achievement). Confounded. Fix: randomize students within one class to use the app or not, blind the evaluator.
2. Without the placebo group we can't separate the drug's real effect from natural improvement plus placebo effect. The extra bit is what we attribute to the drug.